

Variable and Absorption Costing Practice Problem

Summit Sports Manufacturing

Summit Sports Manufacturing manufactures a single product, the **Trail Pack**. The company began operations on January 1 of Year 1 and therefore had **no beginning inventory**.

Cost Information

The following cost information applies in both Year 1 and Year 2:

Item	Amount
Selling price per unit	\$100
Direct materials per unit	\$24
Direct labor per unit	\$16
Variable manufacturing overhead per unit	\$10
Variable selling expense per unit sold	\$6
Annual fixed manufacturing overhead	\$240,000
Annual fixed selling & administrative expenses	\$150,000

Production and Sales Information

	Year 1	Year 2
Units produced	12,000	10,000
Units sold	10,000	12,000

Assume there is **no ending inventory after Year 2**.

Requirements

For **each year**:

1. Calculate the unit product cost using variable costing.
2. Calculate the unit product cost using absorption costing.
3. Prepare an income statement using variable costing.
4. Prepare an income statement using absorption costing.
5. Calculate the amount of fixed manufacturing overhead deferred in or released from inventory.
6. Reconcile the difference between variable-costing operating income and absorption-costing operating income.

For the entire two-year period:

7. Calculate total operating income under variable costing.
8. Calculate total operating income under absorption costing.
9. Explain why cumulative operating income is the same under both methods after all units have been sold.

Note: Calculate the fixed manufacturing overhead rate separately for each year based on that year's production.